

CS-338 Intro to the Theory of Computation

- Lecture #7

➤ **Non Context Free Languages**

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Overview

Last time:

- CFGs and PDA equivalence

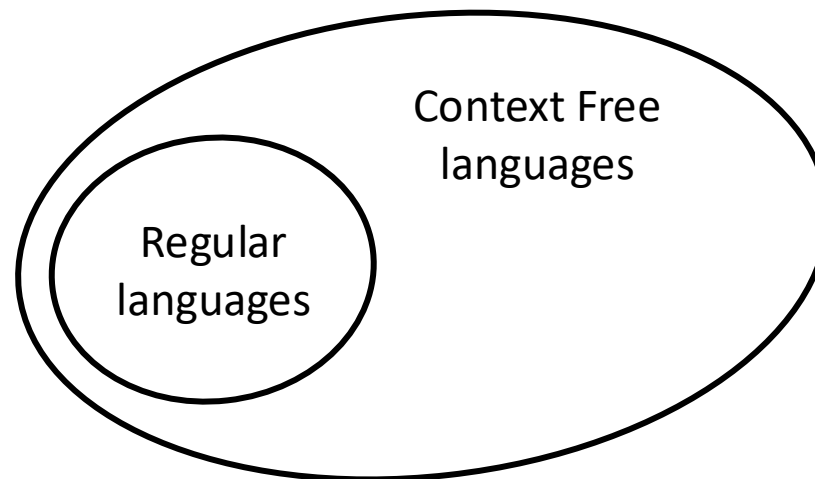
Today: (Sipser §2.3, §3.1)

- Non Context Free Languages
- Turing machines

Recap

Corollary: Every regular language is context free.

	Recognizer	Generator
Regular language	DFA or NFA	Regular expression
Context Free language	PDA	Context Free Grammar



Equivalence of CFGs and PDAs

Recall Theorem: A is a CFL iff some PDA recognizes A

→ Done.

← Done

Corollaries:

- 1) Every regular language is a CFL.
- 2) If A is a CFL and B is regular then $A \cap B$ is a CFL.

Proof sketch of (2):

While reading the input, the finite control of the PDA for A simulates the DFA for B (exercise).

Note 1: If A and B are CFLs then $A \cap B$ may not be a CFL (will show today).
Therefore the class of CFLs is not closed under $\cap$.

Note 2: The class of CFLs is closed under $\cup, \circ, *$

Proving languages not Context Free

$A = \{0^k 1^k 2^l \mid k, l \geq 0\}$ (equal #s of 0s and 1s) CFL?

Let $B = \{0^k 1^k 2^k \mid k \geq 0\}$. CFL?

We will show that B isn't a CFL.

Pumping Lemma for CFLs: For every CFL A , there is a p such that if $s \in A$ and $|s| \geq p$ then $s = uvxyz$ where

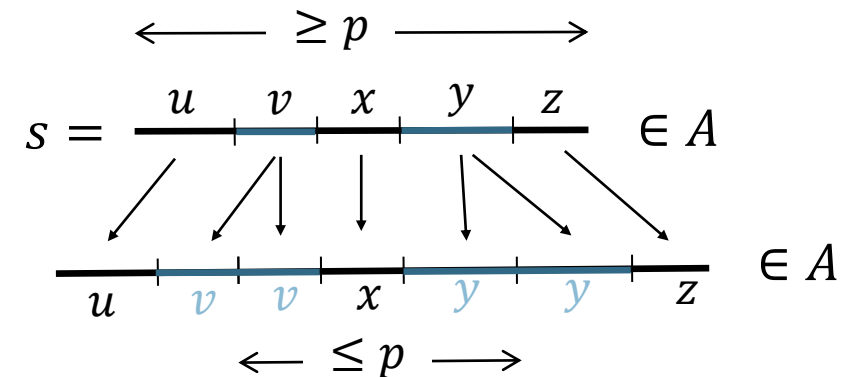
1) $uv^i xy^i z \in A$ for all $i \geq 0$

2) $vy \neq \varepsilon$

3) $|vxy| \leq p$

Informally: All long strings in A are pumpable and stay in A .

- Condition 2 says that either v or y is not the empty string;
- Condition 3 says that vxy has length at most p

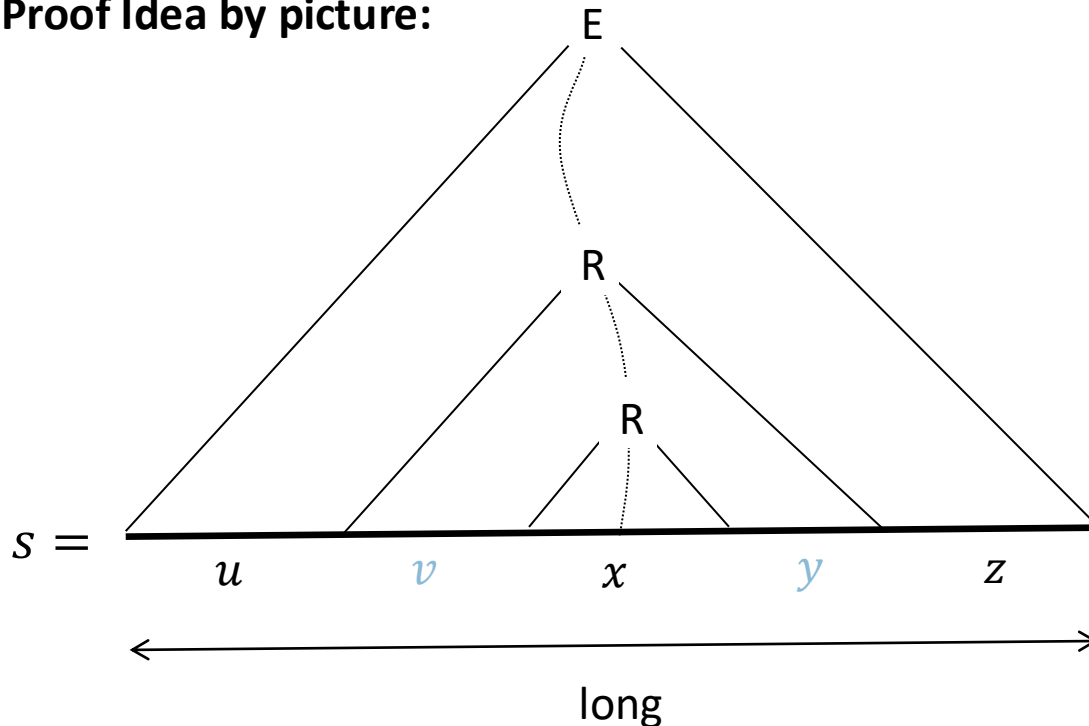


Pumping Lemma – Proof Idea

Pumping Lemma for CFLs: For every CFL A , there is a p such that if $s \in A$ and $|s| \geq p$ then $s = uvxyz$ where

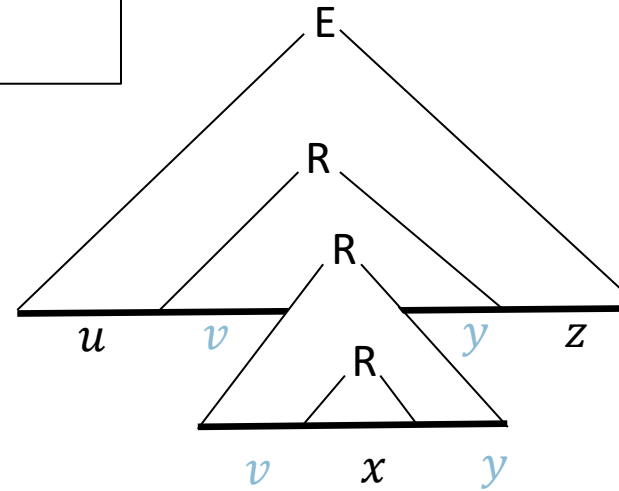
- 1) $uv^i xy^i z \in A$ for all $i \geq 0$
- 2) $vy \neq \varepsilon$
- 3) $|vxy| \leq p$

Proof Idea by picture:

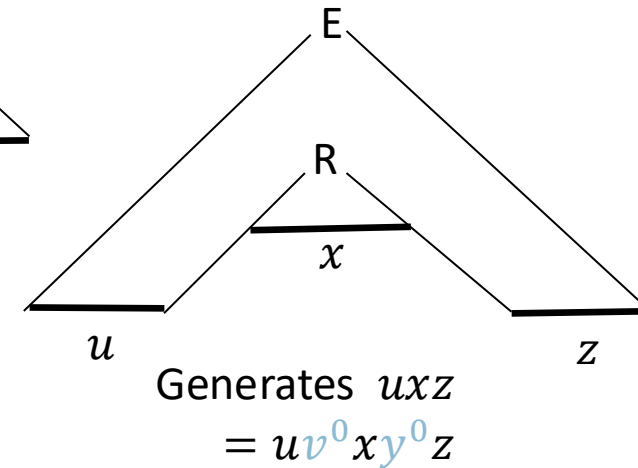


tall

Long $s \rightarrow$
tall parse tree



Generates $uvvxyyz$
 $= uv^2xy^2z$



Generates uxz
 $= uv^0xy^0z$

“cutting and pasting” argument

Pumping Lemma – Proof

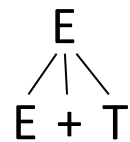
For $s \in A$ where $|s| \geq p$, we have $s = uvxyz$ where:

- 1) $uv^i xy^i z \in A$ for all $i \geq 0$
- 2) $vy \neq \varepsilon$
- 3) $|vxy| \leq p$

Let b = the length of the longest right hand side of a rule (eg., $E \rightarrow E+T$)

= the max branching of the parse tree

Let h = the height of the parse tree for s .



A tree of height h and max branching b has at most b^h leaves.

So it generates a string of length $|s| \leq b^h$ (1).

$\Leftrightarrow$ if $|s| \geq b^{h+1}$ then the height of the parse tree must be at least $h + 1$

$\Rightarrow$ If $|s| \geq b^{|V|+1}$, then $h > |V|$ as desired. (2)

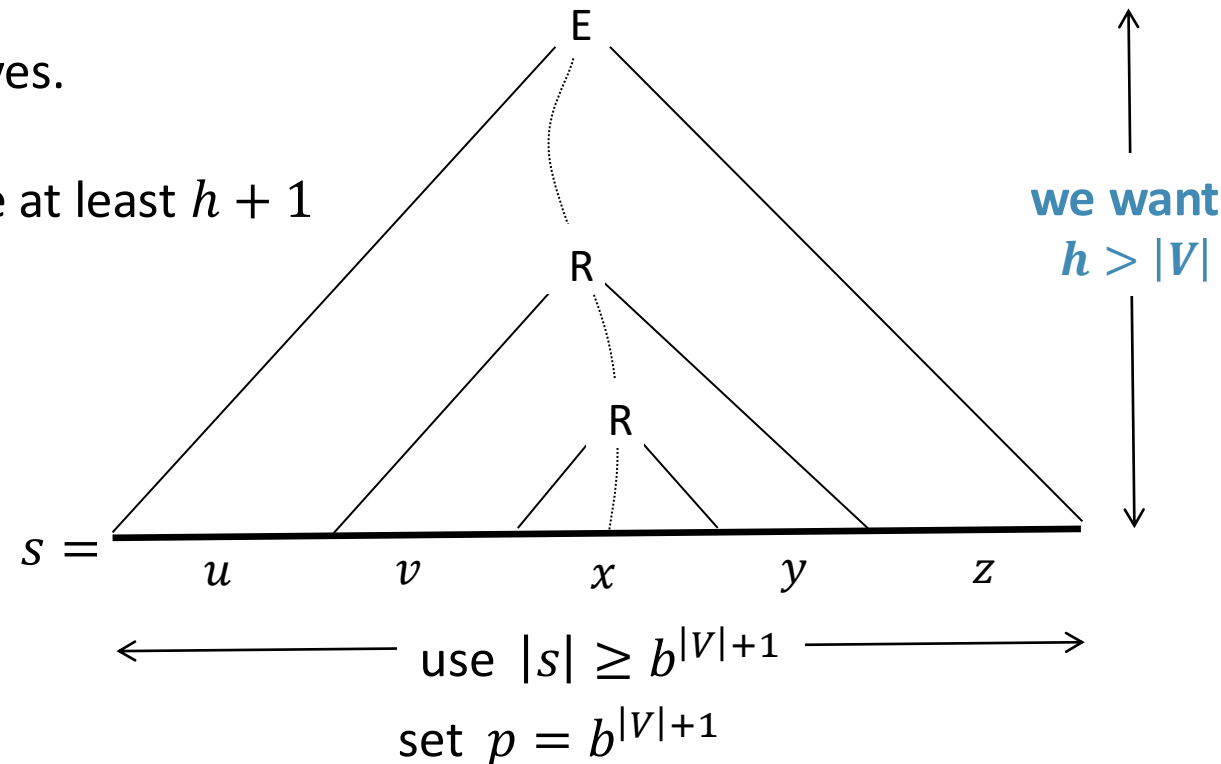
Let $p = b^{|V|+1}$ where $|V|$ = # variables in the grammar.

So if $|s| \geq p > b^{|V|}$.

Then, by (1) & (2) $h > |V|$.

Thus at least $|V| + 1$ variables occur in the longest path.

So some variable R must repeat on a path.



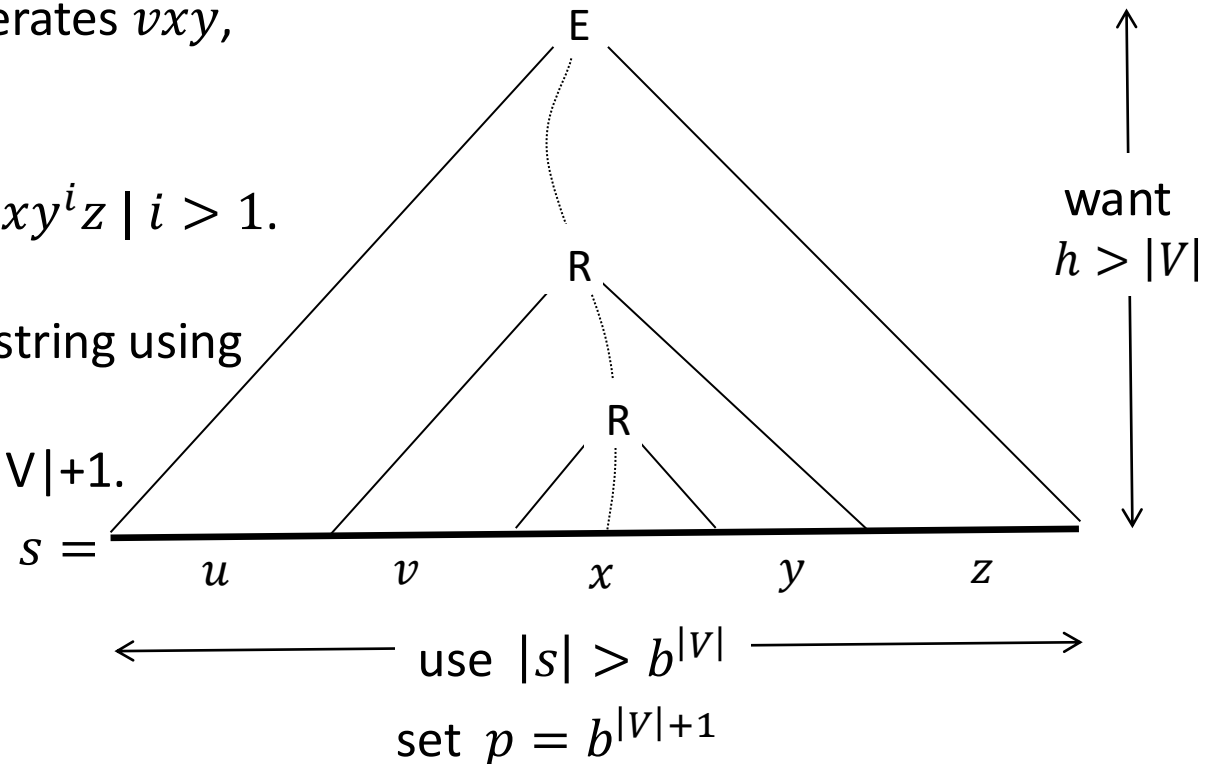
Pumping Lemma – Proof (2)

For $s \in A$ where $|s| \geq p$, we have $s = uvxyz$ where:

- 1) $uv^i xy^i z \in A$ for all $i \geq 0$...cutting and pasting
- 2) $vy \neq \varepsilon$...start with the smallest parse tree for s
- 3) $|vxy| \leq p$...pick the lowest repetition of a variable

- Let T be the parse tree with **smallest** number of nodes to generate s .
- At least $|V| + 1$ variables occur in the longest path.
- So some variable R must repeat on a path: choose the two **lowest**.
- We divide s into $uvxyz$: the upper occurrence of R generates vxy , the lower generates x .
- Condition:

- Replacing the smaller with the larger triangle gives $uv^i xy^i z \mid i > 1$.
Replacing the larger with the smaller gives uxz ($i = 0$).
- If v and y were empty, we could have generated the string using fewer nodes (but T was chosen **smallest** possible)
- The two R variables were chosen amongst the **lowest** $|V| + 1$.
Thus, the subtree generating vxy is high $\leq |V| + 1$.
A tree of this height can generate a string of length at most $s' = vxy \leq b^h = b^{|V|+1} = p$.



Example 1 of Proving Non-CF

Pumping Lemma for CFLs: For every CFL A , there is a p such that if $s \in A$ and $|s| \geq p$ then $s = uvxyz$ where

- 1) $uv^i xy^i z \in A$ for all $i \geq 0$
- 2) $vy \neq \varepsilon$
- 3) $|vxy| \leq p$

Let $B = \{0^k 1^k 2^k \mid k \geq 0\}$

Show: B is not a CFL

Proof by Contradiction:

Assume (to get a contradiction) that B is a CFL.

The CFL pumping lemma gives p as above. Let $s = 0^p 1^p 2^p \in B$.

Pumping lemma says that can divide $s = uvxyz$ satisfying the 3 conditions.

Condition 3 ($|vxy| \leq p$) implies that vxy cannot contain both 0s and 2s.

So $uv^2 xy^2 z$ has unequal numbers of 0s, 1s, and 2s.

Thus $uv^2 xy^2 z \notin B$, violating Condition 1. Contradiction!

Therefore our assumption (B is a CFL) is false. We conclude that B is not a CFL.

$$s = \begin{array}{ccccccc} 0 & 0 & \cdots & 0 & 0 & 1 & 1 & \cdots & 1 & 1 & 2 & 2 & \cdots & 2 & 2 \\ \hline & u & & v & & x & & y & & & & & & z \\ & & & \leftarrow & & \leq p & & \rightarrow & & & & & & \end{array}$$

Example 1 of Proving Non-CF

Pumping Lemma for CFLs: For every CFL A , there is a p such that if $s \in A$ and $|s| \geq p$ then $s = uvxyz$ where

- 1) $uv^i xy^i z \in A$ for all $i \geq 0$
- 2) $vy \neq \varepsilon$
- 3) $|vxy| \leq p$

Let $C = \{0^i 1^j 2^k \mid 0 \leq i \leq j \leq k\}$

Show: C is not a CFL

Proof by Contradiction:

Assume (to get a contradiction) that C is a CFL.

The CFL pumping lemma gives p as above. Let $s = 0^p 1^p 2^p \in C$.

Pumping lemma says that can divide $s = uvxyz$ satisfying the 3 conditions.

Now C contains different numbers of 0s, 1s and 2s: we have to be more careful!

1. v and y contain one type of symbol
 - 0s don't appear $\Rightarrow uv^0 xy^0 z = uxz$ contains same 0s as s but fewer 1s and/or 2s $\Rightarrow \notin C$
 - 1s don't appear $\Rightarrow uv^2 xy^2 z$ either contains more 0s than 1s, or $uv^0 xy^0 z$ more 2s than 1s $\Rightarrow \notin C$
 - 2s don't appear $\Rightarrow uv^2 xy^2 z$ contains more 0s or more 1s than 2s $\Rightarrow \notin C$
2. Either v or y contain more than one symbol
 - $uv^2 xy^2 z$ will not contain the symbols in the correct order $\Rightarrow \notin C$ We conclude that B is not a CFL.

Example 3 of Proving Non-CF

Pumping Lemma for CFLs: For every CFL A , there is a p such that if $s \in A$ and $|s| \geq p$ then $s = uvxyz$ where

- 1) $uv^i xy^i z \in A$ for all $i \geq 0$
- 2) $vy \neq \varepsilon$
- 3) $|vxy| \leq p$

Let $F = \{ww \mid w \in \Sigma^*\}$. $\Sigma = \{0,1\}$.

Show: F is not a CFL.

Assume (for contradiction) that F is a CFL.

The CFL pumping lemma gives p as above. Need to choose $s \in F$. Which s ?

Try $s_1 = 0^p 10^p 1 \in F$. But s_1 can be pumped and stay inside F .

Bad choice of s .

Try $s_2 = 0^p 1^p 0^p 1^p \in F$.

Show s_2 cannot be pumped $s_2 = uvxyz$ satisfying the 3 conditions.

Condition 3 implies that vxy does not overlap two runs of 0s or two runs of 1s.

Therefore, in uv^2xy^2z , two runs of 0s or two runs of 1s have unequal length.

So $uv^2xy^2z \notin F$ violating Condition 1. Contradiction! Thus F is not a CFL.

$$s_1 = 000 \cdots 001000 \cdots 001$$

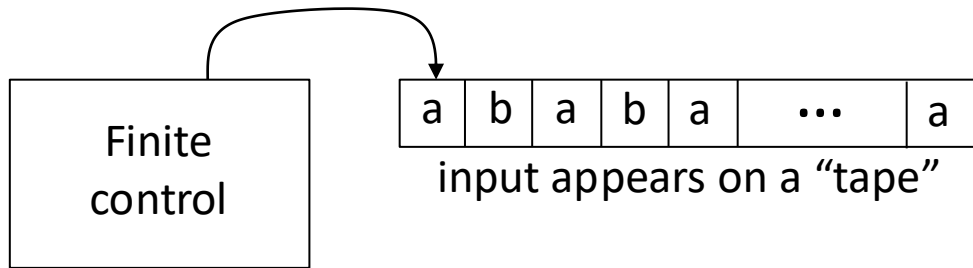
u	$ v $	$ x $	$ y $	z
$\longleftrightarrow \leq p \longleftrightarrow$				

$$s_2 = 0 \cdots 01 \cdots 10 \cdots 01 \cdots 1$$

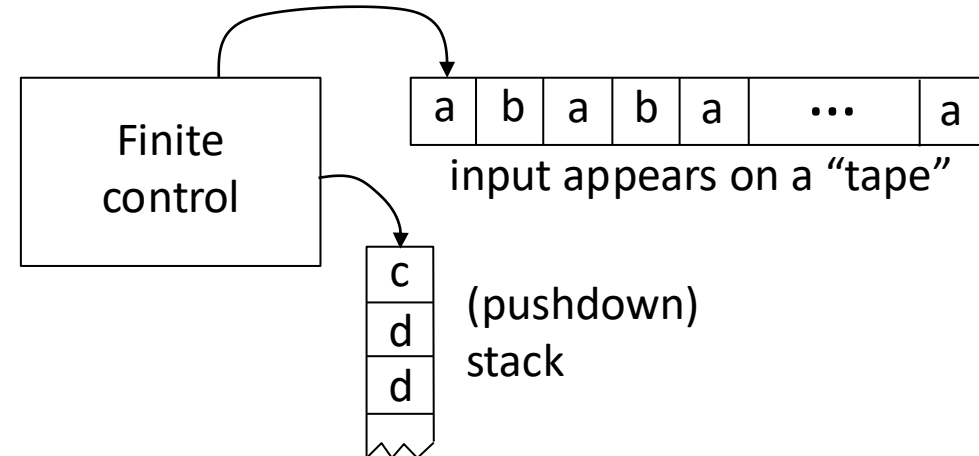
u	$ v $	$ x $	$ y $	z
$\longleftrightarrow$				

Computability Theory: roadmap

- 1) Finite state automata and regular languages ✓
- 2) (Non-deterministic) Pushdown automata and context free languages ✓
- 3) Turing machines



Schematic diagram for DFA



Schematic diagram for PDA